

A New Type of Antiferromagnetic Structure in the Rutile Type Crystal

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(Received March 6, 1959)

A new type of antiferromagnetic spin arrangement, which is theoretically possible in the rutile type crystal and seems to be realized in polianite, MnO_2 , is proposed. This spin arrangement, which will be named the *screw-type structure*, consists of spins screwing along the four-fold crystalline axis. The stability ranges of this and other spin structures are studied, assuming three different exchange interactions. The magnetic properties of the screw-type structure, including the anisotropy energy due to dipolar interaction and that due to crystalline field, the susceptibility, the spin waves, and the neutron diffraction pattern, are studied. Neutron diffraction data for polianite obtained by Erickson (unpublished) can well be accounted for by the screw-type structure with a pitch of $(7/2)c$. The magnetic properties of the MnO_2 crystal are discussed on the basis of this $7/2$ -pitch-structure.

§ 1. Introduction

The purpose of the present paper is to find out possible antiferromagnetic spin arrangements in the rutile type crystal and to study their magnetic properties. In this crystal we expect three different superexchange interactions with exchange integrals J_1 , J_2 and J_3 , the first being the exchange integral between neighboring corner cation and body-center cation, the second that between two cations neighboring along the c -axis, and the third that between two cations neighboring along the a - or b -axis. It will be shown on the basis of a classical energy consideration that, if the first two exchange interactions act in antiferromagnetic way ($J_1 > 0$, $J_2 > 0$), three types of spin superstructure will be possible, the relative stability of which depending on the relative magnitudes of the three exchange integrals. One of them is the MnF_2 -type structure¹⁾, in which the corner spins and body-center spins point antiparallel, the second is the structure suggested by Bizette²⁾, Erickson³⁾ and Yosida⁴⁾, in which the corner spins and the body-center spins form uncorrelated antiferromagnetic sublattices, and the last is a new spin arrangement which we propose and in which positive spins on the corner sites and negative spins on the body-center sites screw along the c -axis. This arrangement will be called the "screw-type" struc-

ture; the pitch of the screw depends on the value of J_2/J_1 .

In the MnF_2 -type spin structure, the exchange coupling between neighbors along the c -axis and that along the a -axis or b -axis act unfavorably, if $J_2 > 0$ and $J_3 > 0$, but the exchange coupling between neighboring corner and body-center cations acts favorably ($J_1 > 0$) and this stabilizes the structure. In the Bizette-Erickson-Yosida-structure, the first two couplings act favorably if $J_2 > 0$, $J_3 > 0$, while the third (J_1) is inactive. Therefore, if J_2 is large compared with J_1 , both being assumed to be positive, and J_3 is small compared with J_1 and J_2 , or J_3 is negative, these two structures will not be stable, and it can be shown that the new screw-type structure becomes more stable when $J_2/J_1 > 1$.

The screw-type structure seems to be realized in polianite, MnO_2 . Bizette and Tsai⁵⁾ (1949) measured the powder susceptibility of polianite and found that its Néel temperature is at 84°K. Bizette²⁾ (1950) measured χ_a and χ_c of a single crystal of it and suggested that spins in this crystal might form two mutually uncorrelated antiferromagnetic sublattices whose spin axes are along $[110]$ and $[\bar{1}\bar{1}0]$, respectively. Erickson³⁾ (1952) carried out a neutron diffraction experiment with a powder sample and found that the corner spins and the body-center spins separately

form antiferromagnetic sublattices. Yosida⁴⁾ (1952) calculated the dipolar anisotropy energy assuming the Bizette-Erickson structure, and qualitatively explained the anisotropy of the susceptibility below and above the Néel temperature. A recent neutron diffraction experiment by Erickson⁶⁾ with a single crystal indicates, however, a different spin structure: the spins are in the c -plane but the magnetic unit cell is seven times as large as the chemical unit cell along the c -axis. It will be shown that the screw-type structure of a pitch of $(7/2)c$ will account for the lines observed by him.

By calculating the classical dipolar anisotropy energy in the screw-type structure, it was found for MnO_2 lattice that when the unit turning angle of the spins, corresponding to the unit c -axis, is less than about 100° , the spins are in an arbitrary plane which contains the c -axis, and that when the unit turning angle is greater than 100° , the spins become perpendicular to the c -axis. It was found further that the crystalline field anisotropy energy stabilizes, to a certain extent, the structures of pitches of simple rational numbers in units of c .

The ratio of the susceptibility in the easy plane to that perpendicular to the easy plane is found to be finite at absolute zero and to depend on the value of J_2/J_1 . For a pitch of $(7/2)c$ a qualitative agreement with the experimental results obtained by Bizette²⁾ is found.

Spin waves in the screw-type structure are

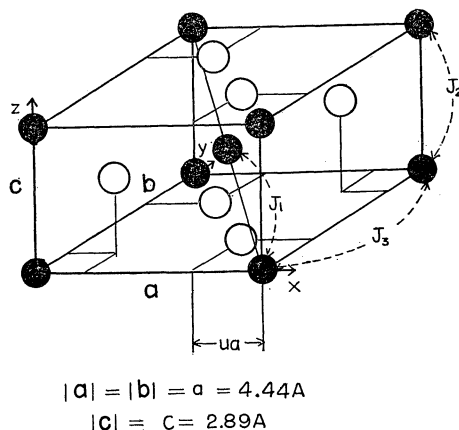


Fig. 1. The rutile type crystal structure of MnO_2 . Solid circles are cations, open circles anions. The value of u is unknown for MnO_2 . For MnF_2 , NiF_2 , $u=0.31$.

also calculated and two non-degenerate branches of the spin wave modes are obtained. It will be shown that there are three modes of antiferromagnetic resonance. Finally, neutron diffraction pattern expected from the screw-type structure will be discussed.

§ 2. Derivation of the Screw-type Structure

As mentioned in the preceding section, we shall assume three different exchange integrals J_1 , J_2 and J_3 (see Fig. 1), and start with a Hamiltonian,

$$H_{\text{ex}} = 2J_1 \sum_{\langle ii' \rangle_1} \mathbf{S}_i \cdot \mathbf{S}_{i'} + 2J_2 \left\{ \sum_{\langle ij \rangle_2} \mathbf{S}_i \cdot \mathbf{S}_j + \sum_{\langle i'j' \rangle_2} \mathbf{S}_{i'} \cdot \mathbf{S}_{j'} \right\} + 2J_3 \left\{ \sum_{\langle ij \rangle_3} \mathbf{S}_i \cdot \mathbf{S}_j + \sum_{\langle i'j' \rangle_3} \mathbf{S}_{i'} \cdot \mathbf{S}_{j'} \right\}, \quad (1)$$

where i and j run over $N/2$ corner cations and i' and j' over $N/2$ body-center cations, $\langle \rangle_1$ means pairs of neighboring corner and body-center cations, $\langle \rangle_2$ pairs of cations neighboring along the c -axis, and $\langle \rangle_3$ those along the a - or b -axis. For a while, we shall treat the spins as classical vectors and look for the minimum of H_{ex} . For this purpose we perform Fourier transformations on $\mathbf{S}_i$ and $\mathbf{S}_{i'}$:

$$\begin{aligned} \mathbf{S}_i &= \sum_{\mathbf{k}} \mathbf{S}(\mathbf{k}) \exp(i\mathbf{k} \cdot \mathbf{r}_i), \quad \mathbf{S}(-\mathbf{k}) = \mathbf{S}^*(\mathbf{k}), \\ \mathbf{S}_{i'} &= \sum_{\mathbf{k}} \mathbf{S}'(\mathbf{k}) \exp(i\mathbf{k} \cdot \mathbf{r}_{i'}), \quad \mathbf{S}'(-\mathbf{k}) = \mathbf{S}'^*(\mathbf{k}), \end{aligned} \quad (2)$$

where $\mathbf{k}$ runs over $N/2$ wave vectors in the Brillouin zone corresponding to the chemical cell. Then, we have

$$\begin{aligned} H_{\text{ex}} \cdot (2/N) &= 16J_1 \sum_{\mathbf{k}} \gamma_{k1} \mathbf{S}(\mathbf{k}) \cdot \mathbf{S}'(-\mathbf{k}) \\ &+ \sum_{\mathbf{k}} (2J_2 \gamma_{k2} + 4J_3 \gamma_{k3}) (\mathbf{S}(\mathbf{k}) \cdot \mathbf{S}(-\mathbf{k}) \\ &+ \mathbf{S}'(\mathbf{k}) \cdot \mathbf{S}'(-\mathbf{k})), \end{aligned}$$

where

$$\begin{aligned} \gamma_{k1} &= \cos(k_x a/2) \cos(k_y a/2) \cos(k_z c/2), \\ \gamma_{k2} &= \cos(k_z c), \\ \gamma_{k3} &= \{\cos(k_x a) + \cos(k_y a)\}/2. \end{aligned} \quad (3)$$

Since $\mathbf{S}_i^2 = \mathbf{S}_{i'}^2 = S^2$, we have

$$\begin{aligned} \sum_{\mathbf{k}, \mathbf{k}'} \mathbf{S}(\mathbf{k}) \cdot \mathbf{S}(\mathbf{k}') \exp\{i(\mathbf{k} + \mathbf{k}') \cdot \mathbf{r}_i\} &= S^2, \\ \sum_{\mathbf{k}, \mathbf{k}'} \mathbf{S}'(\mathbf{k}) \cdot \mathbf{S}'(\mathbf{k}') \exp\{i(\mathbf{k} + \mathbf{k}') \cdot \mathbf{r}_{i'}\} &= S^2. \end{aligned}$$

Under conditions (4), we vary $\mathbf{S}(\mathbf{k})$ and $\mathbf{S}'(\mathbf{k})$ so as to make H_{ex} stationary. Then, equations determining these vectors are found to be as follows:

$$(4J_2\gamma_{k_2} + 8J_3\gamma_{k_3})S(\mathbf{k}) + 16J_1\gamma_{k_1}S'(\mathbf{k}) = \lambda S(\mathbf{k}), \quad (5)$$

$$16J_1\gamma_{k_1}S(\mathbf{k}) + (4J_2\gamma_{k_2} + 8J_3\gamma_{k_3})S'(\mathbf{k}) = \lambda S'(\mathbf{k}), \quad (6)$$

where λ is the Lagrange multiplier which we have assumed to be common to all the N conditions of (4) (for different $\mathbf{r}_i$ and $\mathbf{r}_{i'}$). This assumption means that we are here concerned with a simple sinusoidal change of the spin components in space, with a wave vector $\mathbf{k}$, only distinguishing between the corner and body-center sites, since without this assumption different $\mathbf{k}$ vectors will mix in the equations for $S(\mathbf{k})$ and $S'(\mathbf{k})$. We are therefore imposing only one condition

$$\sum_i S_i^2 + \sum_{i'} S_{i'}^2 = (N/2) \sum_{\mathbf{k}} (S(\mathbf{k}) \cdot S(-\mathbf{k}) + S'(\mathbf{k}) \cdot S'(-\mathbf{k})) = NS^2, \quad (7)$$

in place of N conditions (4). We have thus to test later whether or not the obtained solution will satisfy all the N conditions (4). $(N/2)\lambda S$ is the energy of the total system, H_{ex} , as one can see by multiplying both the sides of (5) by $S(-\mathbf{k})$ and both the sides of (6) by $S'(-\mathbf{k})$ and summing them over $\mathbf{k}^*$.

In order to obtain a non-vanishing solution of (5) and (6), λ must take the following eigenvalues:

$$\lambda = \mp 16J_1\gamma_{k_1} + 4J_2\gamma_{k_2} + 8J_3\gamma_{k_3} \quad (8)$$

The corresponding eigen-vectors satisfy

$$S(\mathbf{k}) = \mp S'(\mathbf{k}), \quad (9)$$

if $\gamma_{k_1} \neq 0$. If $\gamma_{k_1} = 0$, $S(\mathbf{k})$ and $S'(\mathbf{k})$ are uncorrelated to each other. Now (4) can be written as

$$\begin{aligned} S(\mathbf{k})^2 \exp(2i\mathbf{k} \cdot \mathbf{r}_i) + S(-\mathbf{k})^2 \exp(-2i\mathbf{k} \cdot \mathbf{r}_i) \\ + 2S(\mathbf{k}) \cdot S(-\mathbf{k}) = S^2, \\ S'(\mathbf{k})^2 \exp(2i\mathbf{k} \cdot \mathbf{r}_{i'}) + S'(-\mathbf{k})^2 \exp(-2i\mathbf{k} \cdot \mathbf{r}_{i'}) \\ + 2S'(\mathbf{k}) \cdot S'(-\mathbf{k}) = S^2. \end{aligned} \quad (10)$$

In order that the right-hand sides of these equations are independent of $\mathbf{r}_i$ and $\mathbf{r}_{i'}$, either $k_x a$, $k_y a$ and $k_z c$ must be integral multiples of π or $S(\mathbf{k})^2 = S'(\mathbf{k})^2 = 0$. If $k_x a$, $k_y a$ and $k_z c$ are zero and $S(\mathbf{k}) = -S'(\mathbf{k})$, we have the MnF_2 -type antiferromagnetic structure; if $S(\mathbf{k}) = S'(\mathbf{k})$,

we have the ferromagnetic structure. If $k_x a$, $k_y a$ and $k_z c$ are all π , in which case $\gamma_{k_1} = 0$, we have the Bizette-Erickson-Yosida structure. Now, if $S(\mathbf{k})^2 = S'(\mathbf{k})^2 = 0$,

$$S(\mathbf{k}) = \mp S'(\mathbf{k}) = (\mathbf{u}_1 - i\mathbf{u}_2)S/2, \quad (11)$$

where $\mathbf{u}_1$ and $\mathbf{u}_2$ are real unit vectors which are perpendicular to each other. For this solution, (2) becomes

$$\begin{aligned} S_i = \{(\mathbf{u}_1 - i\mathbf{u}_2) \exp(i\mathbf{k} \cdot \mathbf{r}_i) + \text{conj. comp.}\} S/2, \\ = \{\mathbf{u}_1 \cos(\mathbf{k} \cdot \mathbf{r}_i) + \mathbf{u}_2 \sin(\mathbf{k} \cdot \mathbf{r}_i)\} S, \end{aligned} \quad (12)$$

$$S_{i'} = \mp \{\mathbf{u}_1 \cos(\mathbf{k} \cdot \mathbf{r}_{i'}) + \mathbf{u}_2 \sin(\mathbf{k} \cdot \mathbf{r}_{i'})\} S.$$

The corresponding spin structure is such that it can be derived from the MnF_2 -type antiferromagnetic structure (for the $-$ sign in (12)) or from the ferromagnetic structure (for the $+$ sign) by rotating the spins in the $\mathbf{u}_1$, $\mathbf{u}_2$ -plane in such a way that they screw along the direction of $\mathbf{k}$ with a pitch of $2\pi/k$. This may be called the screw-type structure.

The energy of the system, i.e., λ (=total energy $(2/NS^2)$), is given as follows. For the MnF_2 -type antiferromagnetic structure:

$$\lambda = -16J_1 + 4J_2 + 8J_3, \quad (13)$$

for the B. E. Y.-structure:

$$\lambda = -4J_2 - 8J_3, \quad (14)$$

and for the screw-type structure, λ is given by the minimum of (8). This minimum value and the corresponding $\mathbf{k}$ -value are obtained from the equations

$$\begin{aligned} \sin(k_x a/2) \{ \pm J_1 \cos(k_y a/2) \cos(k_z c/2) \\ - J_3 \cos(k_x a/2) \} = 0, \\ \sin(k_y a/2) \{ \pm J_1 \cos(k_z c/2) \cos(k_x a/2) \\ - J_3 \cos(k_y a/2) \} = 0, \\ \sin(k_z c/2) \{ \pm J_1 \cos(k_x a/2) \cos(k_y a/2) \\ - J_2 \cos(k_z c/2) \} = 0. \end{aligned} \quad (15)$$

There are several solutions of these equations, of which the following three are new:

(a) Spins in the same xy -plane are parallel, i.e., $k_x a$ and $k_y a$ are zero, and they screw along the z -axis with a unit screwing angle equal to $k_z c$ (see Fig. 2), where $k_z c$ satisfies.

$$\cos(k_z c/2) = J_1/J_3,$$

$$\text{provided that } |J_2| > |J_1|, \quad (16)$$

(see Fig. 3). The corresponding λ -value is

$$\lambda = -8(J_1^2/J_2) - 4J_2 + 8J_3. \quad (17)$$

(b) Spins in the same yz -plane are parallel and they screw along the x -axis, with a unit screwing angle of $k_x a$, where

* More generally, we can replace λ of the right-hand of (6) by a different Lagrange multiplier λ' , imposing two conditions: $\sum_i S_i^2 = (N/2)S^2$ and $\sum_{i'} S_{i'}^2 = (N/2)S^2$. Then, the energy of the system becomes $(\lambda + \lambda')S^2/2$. λ and λ' are related to each other by the vanishing of the determinant of the equations, and it can be shown that the energy is a minimum when $\lambda = \lambda'$.

$\cos(k_x a/2) = J_1/J_3$, provided that $|J_3| > |J_1|$.

The corresponding λ -value is

$$\lambda = -8(J_1^2/J_3) + 4J_2. \quad (19)$$

The x and y axes may be interchanged.

(c) A three-dimensional screw, with

$$\cos(k_x a/2) = \cos(k_y a/2) = \sqrt{J_2 J_3}/J_1,$$

$$\cos(k_z c/2) = J_3/J_1,$$

$$(|J_1| > \sqrt{J_2 J_3}, J_2 J_3 > 0, |J_3| > |J_1|).$$

The corresponding λ -value is given by

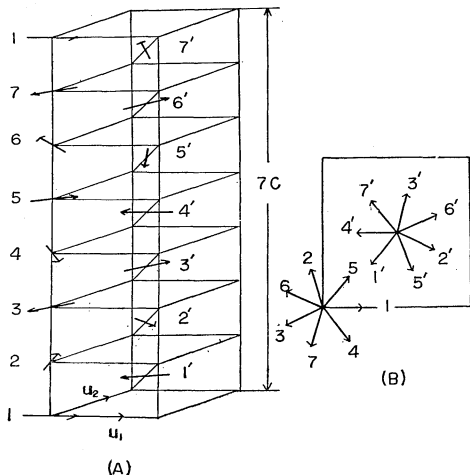


Fig. 2. (A) The screw-type structure with a pitch of $7/2$ (proper screw-type structure). (B) Projection on the c -plane.

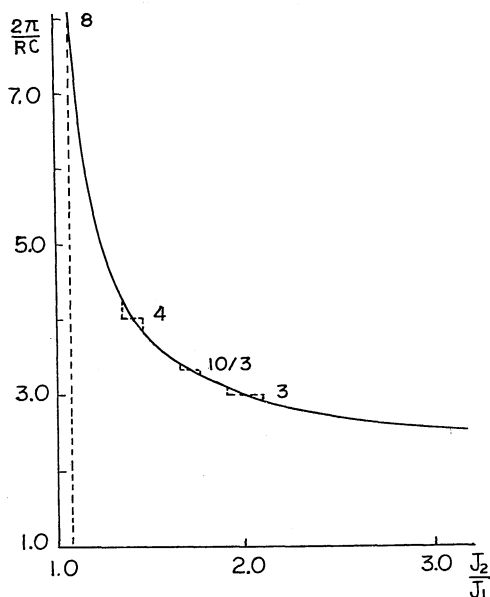


Fig. 3. Relation between the pitch of the screw and J_2/J_1 . Dotted lines indicate (schematically) the modification in certain ranges of J_2/J_1 when the anisotropy energy is considered.

$$\lambda = 8J_2 J_3^2/J_1^2 - 4J_2 - 8J_3. \quad (19)$$

It is interesting to note that the energy λ is the same for both the screwed ferromagnetic ($J_1 < 0$) and screwed MnF_2 -type antiferromagnetic ($J_1 > 0$) structures in the above three cases, as it should be since J_1 appears only as J_1^2 .

Now we compare the energy values of the various structures derived above. Comparing (13) with (14), it can be seen that the boundary of the relative stability of the corresponding two structures is given by

$$J_2 + 2J_3 = 2J_1. \quad (20)$$

If $J_2 + 2J_3 < 2J_1$, the MnF_2 -type structure is more stable, but if $J_2 + 2J_3 > 2J_1$, the B. E. Y.-structure becomes more stable. The structure which screws along the c -axis appears when $J_2 > J_1$ (we assume $J_1 > 0$), and it can be shown that its stability range is bounded by

$$J_2 = J_1 \text{ and } 2J_2 J_3 = J_1^2. \quad (21)$$

The other two screw-type structures have no stability range. Therefore, the whole J_2/J_1 , J_3/J_1 -plane can be divided into four parts as shown in Fig. 4. In this figure, the structure for $J_2 < 0$ and $J_3 > J_1$ (> 0) is such that the spins are parallel to each other along the c -axis and antiparallel along the a and b -axes while the corner spins and body-center spins are uncorrelated.

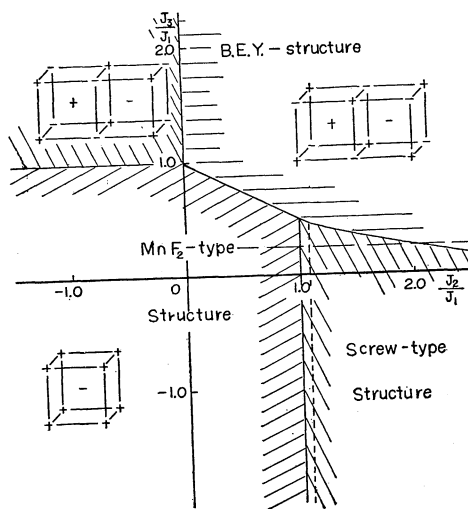


Fig. 4. The stability boundaries of various spin structures in the J_2/J_1 , J_3/J_1 -plane. Dotted vertical line shows the shift of the boundary due to dipolar anisotropy energy (assuming the calculated dipolar interaction in MnO_2 and estimating J_1 , J_2 , J_3 for the same crystal).

In the molecular field approximation, the Néel temperature, T_N , in the screw-type structure with a screw axis along the c -axis, is obtained through a simple calculation as

$$k_B T_N = \{S(S+1)/3\}(8J_1^2/J_2 + 4J_2 - 8J_3), \quad (22)$$

(k_B : the Boltzmann constant) and the paramagnetic Curie temperature, θ_p , as

$$k_B \theta_p = \{S(S+1)/3\}(16J_1 + 4J_2 + 8J_3). \quad (23)$$

Their ratio, θ_p/T_N , is

$$\frac{\theta_p}{T_N} = \frac{4 + (J_2/J_1) + 2(J_3/J_1)}{2(J_2/J_1)^{-1} + (J_2/J_1) - 2(J_3/J_1)}. \quad (24)$$

The maximum allowable value of θ_p/T_N in the screw-type structure of a given pitch occurs at $2(J_3/J_1) = (J_2/J_1)^{-1}$, which is the largest allowable value of $2(J_3/J_1)$ in this structure, and

$$\left(\frac{\theta_p}{T_N}\right)_{\max} = \frac{4 + (J_2/J_1)^{-1} + (J_2/J_1)}{(J_2/J_1)^{-1} + (J_2/J_1)}.$$

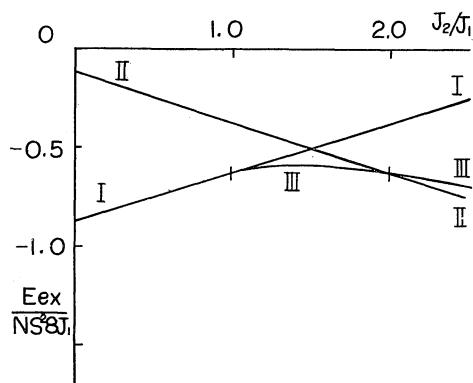


Fig. 5. The energy diagram for the various structures along the dashed line in Fig. 4 ($J_3/J_1 = 1/4$)

- I MnF_2 -type structure,
- II B. E. Y.-structure,
- III Screw-type structure.

The energies of the three superstructures are shown as functions of J_2/J_1 for $J_3/J_1 = 0.25$ in Fig. 5. It is noted that the energy curve of the screw-type structure makes a contact of the second order with that of the MnF_2 -type structure at $J_2/J_1 = 1$.

Since the anisotropy energy has not yet been considered, the spins in the screw-type structure can be parallel to an arbitrary plane. If the spins are parallel to a plane which contains the c -axis, we may call the structure the "cycloidal structure" (Fig. 6). If the spins are perpendicular to the c -axis, i.e., the screw axis, the structure may be called the

"proper screw-type structure". Whether the latter or the former is realized depends on the anisotropy energy, which will be the subject of the next section.

The "proper" $7/2$ -pitch-structure (Fig. 2) is expected as realized in MnO_2 from our analysis of the neutron diffraction data (unpublished) by Erickson⁶⁾. If we assume this, we expect from the conditions of stabilizing the screw-type structure, i.e. from Eq. (16) and $2J_2J_3 < J_1^2$, that in MnO_2 J_2/J_1 is 1.60 and J_3/J_1 smaller than 0.31. On the other hand, we can also determine J_3/J_1 from Eq. (24) and the observed ratio of θ_p (316°K) to T_N (84°K) by Bizette and Tsai⁹⁾, and have $J_3/J_1 \approx 0.56$. This discrepancy will be discussed in section 7. We give here a rough estimate of J_1 for our later purpose assuming somewhat arbitrary $J_3/J_1 = 0.25$: using Eq. (22) and putting $S = 3/2$ and $T_N = 84^\circ\text{K}$, we have $16J_1 \sim 90 \text{ cm}^{-1}$.

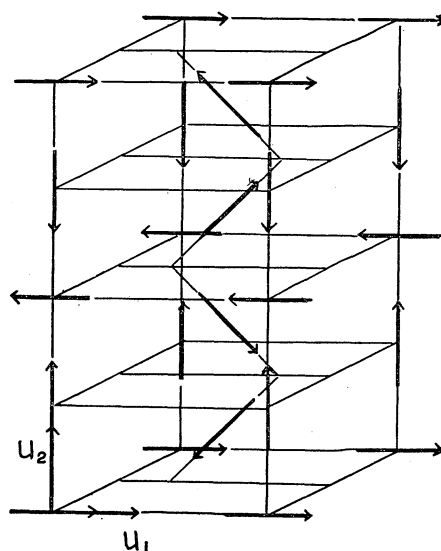


Fig. 6. The cycloidal structure with a pitch of 4.

§ 3. Role of the Anisotropy Energy in the Screw-type Structure

(A) Anisotropy energy due to magnetic dipolar interaction

The magnetic field at each cation due to the magnetic moments of other cations can be expressed as follows:

$$\begin{aligned} H_n^{\text{dip}} &= \sum_l \Phi_{nl} \mu_l + \sum_{l'} \Phi'_{nl} \mu_{l'}, \\ H_{n'}^{\text{dip}} &= \sum_{l'} \Phi_{n'l'} \mu_{l'} + \sum_l \Phi'_{n'l} \mu_l, \end{aligned} \quad (26)$$

where we number layers of corner cations and body-center cations successively along the c -axis, and denote by n and l the layers of corner cations and by n' and l' the layers of body-center cations; $\mathbf{H}_n^{\text{dip}}$ and $\mathbf{H}_{n'}^{\text{dip}}$ therefore mean, respectively, the field at a corner cation on the n th net-plane and the field at a body-center cation on the n' th net-plane, and μ_l and $\mu_{l'}$ are the magnetic moments of cations on the l th net-plane and l' th net-plane. Φ_{nl} and $\Phi_{n'l'}$ are tensors, and evidently

$$\Phi_{nl} = \Phi_{|n-l|}, \quad \Phi_{n'l'} = \Phi'_{|n-l'|}.$$

When the principal axes of the crystal are taken to be the coordinates axes (Fig. 1), it is evident from the crystalline symmetry that Φ_{nl} and $\Phi'_{n'l'}$ are diagonal, and the following relations hold between their principal values;

$$\begin{aligned} \Phi_{|n-l|,x} &= \Phi_{|n-l|,y} = -(1/2)\Phi_{|n-l|,z}, \\ \Phi'_{|n-l'|,x} &= \Phi'_{|n-l'|,y} = -(1/2)\Phi'_{|n-l'|,z}, \end{aligned}$$

Here $\Phi_{|l|,z}$ for instance, is given by

$$\Phi_{|l|,z} = \sum_i r_{li}^{-3} \{3(z_i/r_{li})^2 - 1\},$$

where r_{li} means the distance from a given cation on the zeroth layer to the i th cation on the l th layer, and the summation extends over all i if $l \neq 0$, but it excludes $r_{0i} = 0$ if $l = 0$.

$$\begin{aligned} \mu_l &= g\mu_B \mathbf{S}_l = g\mu_B \langle S_\xi \rangle \{ \mathbf{u}_1 \cos(kz_l) + \mathbf{u}_2 \sin(kz_l) \}, \\ \mu_{l'} &= g\mu_B \mathbf{S}_{l'}, \end{aligned} \quad (27)$$

$$= g\mu_B \langle S_\xi \rangle \{ -\mathbf{u}_1 \cos(kz_{l'}) - \mathbf{u}_2 \sin(kz_{l'}) \},$$

where S_ξ is the component of spin in the direction of the molecular field to which it is subjected and is common to all the spins. We take out two layers which are on the opposite sides of the n th layer at the same distance and combine their μ 's together in the sum of the first equation of (26). Since the sum of these μ 's is a vector which is parallel to the magnetization of the n th layer, we can simplify the calculation of (26). It can be shown that $\mathbf{H}_n^{\text{dip}}$ is expressed as

$$\begin{aligned} \mathbf{H}_n^{\text{dip}} &= (N/2)g\mu_B \langle S_\xi \rangle \varphi_A(\theta) \{ (\mathbf{c}/c)(\mathbf{c}/c) - \mathbf{I} \} \\ &\quad \cdot \{ \mathbf{u}_1 \cos(kz_n) + \mathbf{u}_2 \sin(kz_n) \}, \end{aligned} \quad (28)$$

where $3(\mathbf{c}/c)(\mathbf{c}/c) - \mathbf{I}$ is a dyadic whose diagonal elements are -1 , -1 , 2 , and

$$\varphi_A(\theta) = \{ \varphi(\theta) - \varphi'(\theta) \} / 2, \quad (29)$$

$$\varphi(\theta) = (2/N) \sum_l \Phi_{|l|,z} \cos(kz_l), \quad (30)$$

$$\varphi'(\theta) = (2/N) \sum_{l'} \Phi'_{|l'|,z} \cos(kz_{l'}), \quad (31)$$

where $2\theta (=kc)$ is the unit screwing angle.

$\mathbf{H}_{n'}^{\text{dip}}$ can be expressed in a similar manner. Denoting by n_z the number of net-planes, the dipolar energy per unit volume, E_{dip} , is calculated as follows:

$$\begin{aligned} E_{\text{dip}} &= (N/2n_z) \{ - \sum_n \mu_n \cdot \mathbf{H}_n^{\text{dip}} - \sum_{n'} \mu_{n'} \cdot \mathbf{H}_{n'}^{\text{dip}} \}, \\ &= \{ 1 - (3/2)(u_{1z}^2 + u_{2z}^2) \} \varphi_A(\theta) (N\mu/2)^2, \\ \mu &= g\mu_B \langle S_\xi \rangle. \end{aligned}$$

If $\mathbf{u}_3$ is defined by $\mathbf{u}_3 = \mathbf{u}_1 \times \mathbf{u}_2$ and θ is the angle between $\mathbf{u}_3$ and the c -axis, it follows that

$$u_{1z}^2 + u_{2z}^2 = 1 - u_{3z}^2 = 1 - \cos^2 \theta.$$

Thus

$$E_{\text{dip}} = (1/2)(3 \cos^2 \theta - 1) \varphi_A(\theta) (N\mu/2)^2. \quad (32)$$

The values of $\varphi(\theta)$ and $\varphi'(\theta)$ were calculated

Table I.

θ°	$-\varphi(\theta)$	$-\varphi'(\theta)$	$\varphi_A(\theta)$
0	1.27	11.78	5.26
15	1.91	11.26	4.68
30	3.64	9.83	3.10
45	5.94	7.74	0.90
51.4	6.96	6.72	-0.12
60	8.20	5.29	-1.46
75	9.81	2.68	-3.57
90	10.39	0	-5.20

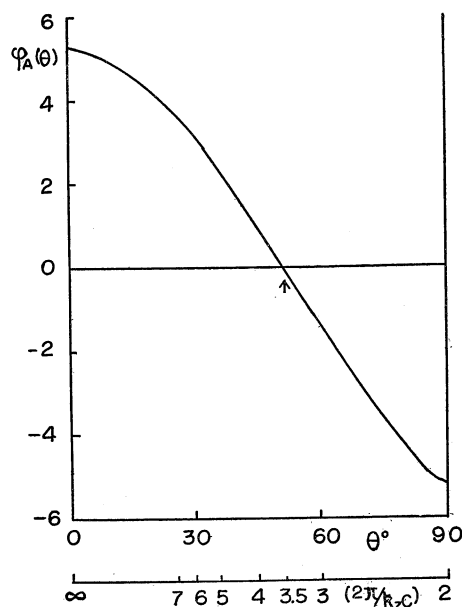


Fig. 7. The variation of $\varphi_A(\theta)$ versus θ in MnO_2 . A small arrow indicates the point corresponding to a pitch of 3.5.

by the direct sum method first summing over cations on each net-plane and then summing over different net-planes. The results are shown in Table I and Fig. 7 for the lattice of MnO_2 . Putting $S=3/2$ and $g=2$, E_{dip}/N becomes

$$E_{\text{dip}}/N = 0.0169 \varphi_A(\theta) (\langle S_x \rangle / S)^2 (3 \cos^2 \theta - 1) \text{cm}^{-1}. \quad (33)$$

If $\varphi_A(\theta)$ is positive, the easy planes of the spins is parallel to the c -axis ($\theta=90^\circ$), i.e., the cycloidal structure becomes stable, but if $\varphi_A(\theta)$ is negative, the proper screw-type structure ($\theta=0^\circ$) becomes stable. In Fig. 7 it is seen that $\varphi_A(\theta)$ changes sign at $\theta=50.8^\circ$. In the $7/2$ -pitch-structure for which $\theta=51.4^\circ$, θ is a little above this value and so the magnetizations are stabilized in the c -plane. E_{dip}/N of the $7/2$ -pitch-structure is

$$E_{\text{dip}}/N = -0.002 (\langle S_x \rangle / S)^2 (3 \cos^2 \theta - 1) \text{cm}^{-1}. \quad (34)$$

This anisotropy energy is very small and may not be important compared with the crystalline field anisotropy energy which will be discussed later.

The dipolar energy of the MnF_2 -type structure, in which spins point parallel to the c -axis, is $-(1/2)\varphi_A(0)(N\mu/2)^2$. This is twice as large as that of the cycloidal structure for vanishing θ . On the other hand, the difference in the exchange energy for the MnF_2 -type and screw-type structures is very small when $J_2/J_1 \sim 1$, as mentioned in section 2. Therefore, if J_2 exceeds J_1 only a little, the MnF_2 -type structure will remain to be more stable than the cycloidal structure, since the corresponding increase in exchange energy will be over-compensated by dipolar energy; in other words, the stability boundary shifts a little towards a larger value of J_2/J_1 . For MnO_2 , the difference in dipolar energy per cation corresponding to these two possible structure, namely $(1/2N)\varphi_A(0)(N\mu/2)^2$, is about 0.09cm^{-1} . In order to estimate the order of magnitude of the shift in this case, we assume $16J_1 \sim 90 \text{cm}^{-1}$ (see the preceding section), then the boundary value of J_2/J_1 turns out to be 1.08 (Fig. 4). The pitch corresponding to this new boundary is about 8, i.e., screw-type structures with larger pitches than about 8 are unstable (Fig. 3). The stability boundary shifts also due to the crystalline field anisotropy energy, which will be treated in the

next subsection.

(B) Anisotropy energy due to crystalline field

The Mn^{4+} ions in polianite may have an anisotropy energy due to the crystalline field. Let us consider the orthorhombic anisotropy energy of the lowest order:

$$\begin{aligned} D' \beta_3^2 - E' \beta_1 \beta_2, \\ \text{for the corner cations,} \\ D' \beta_3^2 + E' \beta_1 \beta_2, \\ \text{for the body-center cations,} \end{aligned} \quad (35)$$

where $\beta_1, \beta_2, \beta_3$ are the direction cosines of the spins referred to the principal axes of the crystal. These energies are derived from the effective spin Hamiltonian,

$$\begin{aligned} \sum_i \{ D S_{iz}^2 - E (S_{ix} S_{iy} + S_{iy} S_{ix}) \} \\ + \sum_{i'} \{ D S_{i'z}^2 + E (S_{i'x} S_{i'y} + S_{i'y} S_{i'x}) \}. \end{aligned} \quad (36)$$

Making use of the molecular field approximation, we have

$$D'/D = E'/E = \{ 3 \langle S_x^2 \rangle - S(S+1) \} / 2. \quad (37)$$

Treating (35) as a classical perturbation to the exchange energy, one can easily obtain the first order crystalline anisotropy energy (per cation) to be

$$(D'/2) \sin^2 \theta, \quad (38)$$

(the terms with E' are averaged out to zero). Hence, for positive D' the proper screw-type and for negative D' the cycloidal structure are stabilized. For negative D' , the stability boundary between the MnF_2 -type structure and screw-type structure shifts in the same way as in the case of dipolar anisotropy energy.

The anisotropy energy in the easy plane of the proper screw-type structure will vary discontinuously as a function of pitch due to E' -terms, if we go over to higher order perturbation calculation. Due to the E' -terms of (35), the corner spins tend towards $[110]$ and the body-center spins towards $[1\bar{1}0]$, assuming $E' > 0$, and it can be shown that a structure with a pitch of a simple rational number is stabilized by this effect. For example, the anisotropy energy in the c -plane of the 6-pitch-structure is expressible as a combination of $\sin 6\theta^*$ and $\cos 6\theta^*$ and that of the 4-pitch-structure as a combination of $\sin 8\theta^*$ and $\cos 8\theta^*$, where θ^* specifies the angle of rotation of the spins as a whole in the c -

plane, the minima of which these anisotropy energies depressing the total energy. The anisotropy energy in the easy plane may also come from the orthorhombic anisotropy energy of a higher order (for example, a higher order quantum-mechanical perturbation of the spin-orbit coupling). In any case, the anisotropy energy in the easy plane must appear at lower orders for simple pitches than for complex pitches. In particular, for irrational pitches we have no anisotropy energy. As a result, the pitch as a function of J_3/J_1 will become as shown schematically in Fig. 3. However, the width of each part of constant pitch in this figure may be very small, as it arises from a higher order perturbational calculation. We find a similar situation in the cycloidal structure considering only the dipolar interaction.

§ 4. Susceptibility

In calculating the susceptibility, we shall

neglect the dipolar interaction, which will be of minor importance compared with the exchange interaction, if the Néel temperature of the substance is high ($\geq 1^\circ\text{K}$). The crystal-line field anisotropy energy will be taken into account through an anisotropic g -factor. By applying an external field, the pitch of the screw will be slightly changed if the field is applied in the plane of the spins, but it can be shown that the change does not occur to first order in $\mathbf{H}$. Therefore, we shall assume that the pitch is a given constant.

When the external field $\mathbf{H}$ is applied perpendicularly to the plane in which the spins lie, we obtain a temperature-independent susceptibility, $\chi_{s\perp}$, as we shall show below. In this case, all the μ_n and $\mu_{n'}$ must make the same angle, τ ($< 90^\circ$), with $\mathbf{H}$. The torque on, say, μ_n must vanish, i.e.

$$\mu_n \times (\mathbf{H}_n^{\text{ex}} + \mathbf{H}) = 0 \quad (39)$$

where

$$\begin{aligned} \mathbf{H}_n^{\text{ex}} &= -(\alpha_1/2)(\mu_{n'} + \mu_{n'-1}) - (\alpha_2/2)(\mu_{n+1} + \mu_{n-1}) - \gamma\mu_n, \\ \alpha_1 &= 16J_1(g\mu_B)^{-2}, \quad \alpha_2 = 4J_2(g\mu_B)^{-2}, \quad \gamma = 8J_3(g\mu_B)^{-2}. \end{aligned} \quad (40)$$

If the z' -axis is taken along $\mathbf{H}$, x' -axis along the original μ_n -direction (for no external field) and the y' -axis perpendicular to the both, we have from the y' -component of (39),

$$\mu \cos \tau (\alpha_1 \mu \sin \tau \cos \theta - \alpha_2 \mu \sin \tau \cos 2\theta) - \mu \sin \tau (-\alpha_1 \mu \cos \tau - \alpha_2 \mu \cos \tau + H) = 0,$$

or

$$\mu \cos \tau = H / \{ \alpha_1(1 + \cos \theta) + \alpha_2(1 - \cos 2\theta) \}.$$

The magnitude of the total magnetization is

$$\begin{aligned} N\mu_{z'} &= N\mu \cos \tau = H / \{ A_1(1 + \cos \theta) + A_2(1 - \cos 2\theta) \}, \\ A_1 &= \alpha_1/N, \quad A_2 = \alpha_2/N \end{aligned}$$

so that the susceptibility is given by

$$\chi_{s\perp} = 1 / \{ A_1(1 + \cos \theta) + A_2(1 - \cos 2\theta) \}. \quad (41)$$

It might be reminded that $\cos \theta = J_1/J_2 = A_1/4A_2$.

When $\mathbf{H}$ is applied in the spin plane, we have a different susceptibility, $\chi_{s//}$. In this case, we denote the angle between the equilibrium direction of μ_n in the field and $\mathbf{H}$ itself by θ_n and similarly the angle between $-\mu_{n'}$ and $\mathbf{H}$ by $\theta_{n'}$. Then we have

$$\begin{aligned} |\mu_n \times (\mathbf{H}_n^{\text{ex}} + \mathbf{H})| &= -\alpha_1 \mu_n \{ \mu_{n'} \sin(\theta_{n'} - \theta_n) \\ &\quad - \mu_{n'-1} \sin(\theta_n - \theta_{n'-1}) \} + \alpha_2 \mu_n \{ \mu_{n+1} \sin(\theta_{n+1} - \theta_n) \\ &\quad - \mu_{n-1} \sin(\theta_n - \theta_{n-1}) \} - \mu_n H \sin \theta_n = 0, \end{aligned} \quad (42)$$

and a similar equation for $\mu_{n'}$. Here $\mathbf{H}_n^{\text{ex}}$, given by (40), is different from that without external field since μ_n and $\mu_{n'}$ have changed, not only as regards their directions but also as

regards their lengths. The latter is determined by

$$\mu_n = g\mu_B \langle S_{\xi} \rangle = g\mu_B S B_S(y_n), \quad y_n = (g\mu_B S / k_B T) |\mathbf{H}_n^{\text{ex}} + \mathbf{H}|, \quad (43)$$

and similar equations for $\mu_{n'}$, where $B_S(y)$ is the Brillouin function. When no external field is present, $\theta_{n+1}^{(0)} - \theta_n^{(0)}$ and $\theta_{n'}^{(0)} - \theta_n^{(0)}$ are given by

$$\theta_{n+1}^{(0)} - \theta_n^{(0)} = 2\theta, \quad \theta_{n'}^{(0)} - \theta_n^{(0)} = \theta, \quad (44)$$

and all the μ_n and $\mu_{n'}$ are equal to

$$\mu = g\mu_B S B_S(y_0), \quad y_0 = (g\mu_B S / k_B T) \mu (\alpha_1 \cos \theta - \alpha_2 \cos 2\theta - \gamma).$$

If angular shifts, $\Delta\theta_n = \theta_n - \theta_n^{(0)}$, $\Delta\theta_{n'} = \theta_{n'} - \theta_{n'}^{(0)}$, and variations of μ_n and $\mu_{n'}$, denoted as $\mu\epsilon_n$ and $\mu\epsilon_{n'}$, due to the external field are considered to the first order in H , Eqs. (42) and (43) can be linearized, and the following difference equations are obtained for ϵ_n , $\epsilon_{n'}$ and $\Delta\theta_n$, $\Delta\theta_{n'}$:

$$\begin{aligned} & (\alpha_1/2)(\epsilon_{n'} - \epsilon_{n'-1}) \sin \theta - (\alpha_2/2)(\epsilon_{n+1} - \epsilon_{n-1}) \sin 2\theta \\ & + (\alpha_1/2)(\Delta\theta_{n'} + \Delta\theta_{n'-1} - 2\Delta\theta_n) \cos \theta - (\alpha_2/2)(\Delta\theta_{n+1} + \Delta\theta_{n-1} - 2\Delta\theta_n) \cos 2\theta \\ & - (H/\mu) \sin (2n\theta + \theta_0^{(0)}) = 0, \end{aligned} \quad (45)$$

$$\begin{aligned} & -(\alpha_1/2)(\epsilon_{n'} + \epsilon_{n'-1}) \cos \theta + \{t\epsilon_n + (\alpha_2/2)(\epsilon_{n+1} + \epsilon_{n-1}) \cos 2\theta\} \\ & - (\alpha_1/2)(\Delta\theta_n - \Delta\theta_{n-1}) \sin \theta + (\alpha_2/2)(\Delta\theta_{n+1} - \Delta\theta_{n-1}) \sin 2\theta \\ & + (H/\mu) \cos (2n\theta + \theta_0^{(0)}) = 0, \end{aligned} \quad (46)$$

where

$$t = k_B T \{(g\mu_B)^2 B_S'(y_0)\}^{-1} + \gamma$$

The following solutions are easily found from these difference equations:

$$\begin{aligned} \Delta\theta_n &= -d_1 \sin (2n\theta + \theta_0^{(0)}), & \Delta\theta_{n'} &= d_1 \sin \{(2n+1)\theta + \theta_0^{(0)}\}, \\ \epsilon_n &= d_2 \cos (2n\theta + \theta_0^{(0)}), & \epsilon_{n'} &= -d_2 \cos \{(2n+1)\theta + \theta_0^{(0)}\}, \end{aligned} \quad (47)$$

where

$$\begin{aligned} Nd_1 &= (H/\mu)(V - W)/(UV - W^2), & Nd_2 &= (H/\mu)(U - W)(UV - W^2), \\ U &= (t/N) + A_1 \cos^2 \theta + A_2 \cos^2 2\theta, & V &= A_1 \cos \theta (1 + \cos \theta) - A_2 \cos 2\theta (1 - \cos 2\theta), \\ W &= A_1 \sin^2 \theta + A_2 \sin^2 2\theta, \end{aligned}$$

The magnitude of the total magnetization is

$$(N/n_z) \left\{ \sum_n \mu_n \cos \theta_n - \sum_{n'} \mu_{n'} \cos \theta_{n'} \right\} = (N\mu/2)(d_1 + d_2),$$

so that the susceptibility is given by

$$\chi_{s//} = (1/2)(U + V - 2W)/(UV - W^2). \quad (48)$$

This $\chi_{s//}$ has a finite value at absolute zero. For $T \rightarrow 0$, t becomes infinite and so $U \rightarrow \infty$, so that

$$(\chi_{s//})_{T=0} = 1/(2V) = (1/2) \{A_1 \cos \theta (1 + \cos \theta) - A_2 \cos 2\theta (1 - \cos 2\theta)\}^{-1}. \quad (49)$$

The ratio of $\chi_{s//}$ to $\chi_{s\perp}$ at absolute zero is a function of θ only (see Fig. 8), and is given by

$$\begin{aligned} \langle \chi_{s//} \rangle_{T=0} / \chi_{s\perp} &= (1/2) \{ (1 + \cos \theta) + (A_2/A_1)(1 - \cos 2\theta) \} \\ &\times \{ \cos \theta (1 + \cos \theta) - (A_2/A_1) \cos 2\theta (1 - \cos 2\theta) \}^{-1}, \end{aligned} \quad (50)$$

where

$$\cos \theta = A_1 / (4A_2).$$

$\chi_{s//}$ increases monotonically with rising temperature up to T_N . At $T = T_N$, t becomes

$$(t/N)_{T=T_N} = A_1 \cos \theta - A_2 \cos 2\theta,$$

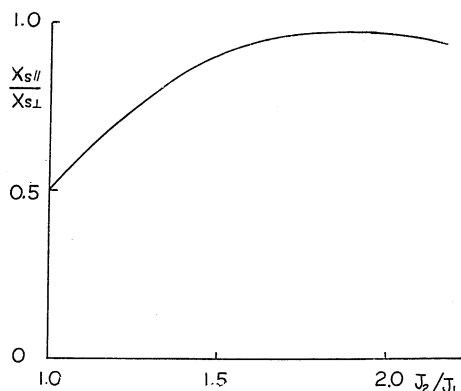


Fig. 8. Variation of $\chi_{s//}/\chi_{s\perp}$ against J_2/J_1 .

and it can be shown that $\chi_{s//}$ coincides with $\chi_{s\perp}$.

When the anisotropy of the g -factor and the anisotropic crystalline field are taken into consideration, Eq. (39) will be replaced by

$$\mu_n \times (\mathbf{H}_n^{\text{ex}} + \mathbf{H}_n^{\text{c.a.}} + \mathbf{H}^*) = 0,$$

and y_n in Eq. (43) by

$$y_n = (g\mu_B S / k_B T) |\mathbf{H}_n^{\text{ex}} + \mathbf{H}_n^{\text{c.a.}} + \mathbf{H}^*|.$$

Here $\mathbf{H}_n^{\text{c.a.}}$ (effective field due to crystalline anisotropy) is given, corresponding to (35), by

$$\mathbf{H}_n^{\text{c.a.}} = -(D'/\mu) \beta_{3n}(\mathbf{c}/c) - (E'/\mu) (\beta_{2n}\mathbf{a}/a + \beta_{1n}\mathbf{b}/a),$$

and $\mathbf{H}^*$, $\mathbf{H}^{*'}$ are defined by

$$\mathbf{g}^2 \mathbf{H} = g^2 \mathbf{H}^*, \quad \mathbf{g}'^2 \mathbf{H} = g^2 \mathbf{H}^{*'},$$

where $\mathbf{g}$ and $\mathbf{g}'$ are g -tensors for the corner and body-center cations, respectively. Their principal values will be denoted as g_1, g_2, g_3 corresponding to the axes $[110], [\bar{1}\bar{1}0], [001]$ for the corner sites and $[1\bar{1}0], [\bar{1}\bar{1}0], [001]$ for the body-center sites. Assuming the proper screw-type structure, we obtain, in place of $\chi_{s\perp}$ and $\chi_{s//}$,

$$\chi_c = (g_3/g)^2 \{ A_1(1 + \cos \theta) + A_2(1 - \cos 2\theta) + (D'/\mu^2)(2/N) \}^{-1}, \quad (51)$$

$$\chi_a = (1/2) \{ (g_1^2 + g_2^2) / 2g^2 \} (U + V - 2W) / (UV - W^2). \quad (52)$$

At absolute zero, χ_a reduces to

$$(\chi_a)_{T=0} = (1/2) \{ (g_1^2 + g_2^2) / 2g^2 \} (1/V). \quad (53)$$

We shall calculate the principal susceptibilities for the proper screw-type structure of pitch $7/2$ and compare them with those measured by Bizette²⁾ with a single crystal of MnO_2 . As $4A_2/A_1 = J_2/J_1 = 1.60$ for this pitch, we obtain $(\chi_{s//})_{T=0} / \chi_{s\perp} = 0.94$ from (50), neglecting the anisotropy of the g -factor and $\mathbf{H}_n^{\text{c.a.}}$. However, the experimental value of $(\chi_a)_{T=0} / (\chi_c)_{T=0}$ is 0.79. Furthermore, χ_a is smaller than χ_c above the Néel temperature according to Bizette. These discrepancies may perhaps be explained by the anisotropy of the g -factor, crystalline anisotropy field and dipolar interaction.

It was already pointed out by Yosida⁴⁾ that an observed value of χ_a/χ_c larger than $1/2$ at absolute zero can be qualitatively accounted by the B.E.Y.-structure and the anisotropic property of the susceptibility above the Néel temperature by dipolar interaction. However, the dipolar interaction does not seem to be sufficient to account for the latter quantitatively.

If we assume $g_1/g_3 \approx 0.84$ (putting $g_1 = g_2$ for simplicity and neglecting D' in expression (51) for χ_c), we can account for the difference between 0.94 and 0.79 for $(\chi_{s//})_{T=0} / \chi_{s\perp}$. D is related with g -values by

$$D \sim \lambda(g_3 - g_1)/2,$$

where λ is the spin-orbit coupling constant and is 140 cm^{-1} for the free Mn^{4+} . Since $g_3 > g_1$, D is positive, which means that the crystalline anisotropy energy, in addition to the dipolar energy, stabilizes the proper screw-type structure. This is consistent with our assumption made at the beginning. If we adopt 1.49 for the value of $\{(2g_1^2 + g_3^2)/3\}^{1/2}$, as Bizette and Tsai⁵⁾ found from their powder susceptibility measurements above the Néel temperature assuming $S=3/2$ for Mn^{4+} , we find $D \sim 9 \text{ cm}^{-1}$. This is somewhat too large to be expected from data on other salts and may not be reliable, but the positive sign of D will be certain.

Finally, it may be noted that, when a strong field is applied in the c -plane for the proper screw-type structure, we expect flopping of the spin direction as in the usual antiferromagnet. Namely, above a certain critical field, spins as a whole will turn and become perpendicular to the applied field, in other words, the proper screw-type structure will change to a cycloidal structure. This critical field can easily be derived from (33) and (38), as

$$H_c^2 = \{ND' - 3\varphi_A(\theta)(N\mu/2)^2\} / (\chi_{s\perp} - \chi_{s\parallel}) .$$

H_c is about 10^6 oe at absolute zero for the above value of D ($D' = 3D/2$ for $S=3/2$).

§ 5. Spin Waves and Antiferromagnetic Resonance in the Proper Screw-type Structure

We start with the exchange Hamiltonian (1). First, we perform an orthogonal coordinate transformation for spin components which is different for different positions of the spin:

$$\begin{aligned} S_{ix} &= S_{i\xi} \cos(kz_i) - S_{i\eta} \sin(kz_i), & S_{iy} &= S_{i\xi} \sin(kz_i) + S_{i\eta} \cos(kz_i), \\ S_{i'x} &= -S_{i'\xi'} \cos(kz_{i'}) + S_{i'\eta'} \sin(kz_{i'}), & S_{i'y} &= -S_{i'\xi'} \sin(kz_{i'}) - S_{i'\eta'} \cos(kz_{i'}) . \end{aligned} \quad (54)$$

ξ and ξ' are the equilibrium directions of the spins at the corner sites and body-center sites, respectively. Quantizing the spins along ξ and ξ' and using Holstein-Primakoff's technique⁶⁾, we can derive the following spin-wave Hamiltonian:

$$\begin{aligned} H_{sw} &= -N8J_1S^2 \cos \theta + N2J_2S^2 \cos 2\theta + N4J_3S^2 \\ &\quad + 2J_1S \sum_{\langle ii' \rangle_1} [(a_i^* a_i + b_{i'}^* b_{i'}) \cos \theta + (a_i^* b_{i'} + a_i b_{i'}^*) \\ &\quad \times (1 - \cos \theta)/2 - (a_i b_{i'} + a_i^* b_{i'}^*) (1 + \cos \theta)/2] \\ &\quad + 2J_2S \left[\sum_{\langle ij \rangle_2} \{ -(a_i^* a_i + a_j^* a_j) \cos 2\theta + (a_i^* a_j + a_i a_j^*) \right. \\ &\quad \times (1 + \cos 2\theta)/2 - (a_i a_j + a_i^* a_j^*) (1 - \cos 2\theta)/2 \} \\ &\quad + \sum_{\langle i'j' \rangle_2} \{ -(b_{i'}^* b_{i'} + b_{j'}^* b_{j'}) \cos 2\theta + (b_{i'}^* b_{j'} + b_{i'} b_{j'}^*) \\ &\quad \times (1 + \cos 2\theta)/2 - (b_{i'} b_{j'} + b_{i'}^* b_{j'}^*) (1 - \cos 2\theta)/2 \} \\ &\quad + 2J_3S \left[\sum_{\langle ij \rangle_3} \{ -(a_i^* a_i + a_j^* a_j) + (a_i a_j^* + a_i^* a_j) \} \right. \\ &\quad \left. + \sum_{\langle i'j' \rangle_3} \{ -(b_{i'}^* b_{i'} + b_{j'}^* b_{j'}) + (b_{i'} b_{j'}^* + b_{i'}^* b_{j'}) \} \right] , \end{aligned}$$

where a_i , $b_{i'}$, a_i^* , $b_{i'}^*$ are the annihilation and creation operators and are connected with the spin operators by the following equations:

$$\begin{aligned} S_{i\xi} &= S - a_i^* a_i, & S_{i\eta} + iS_{iz} &= (2S)^{1/2} \{ 1 - (a_i^* a_i / 2S) \}^{1/2} a_i \approx (2S)^{1/2} a_i, \\ S_{i'\xi'} &= S - b_{i'}^* b_{i'}, & S_{i'\eta'} + iS_{i'z} &\approx (2S)^{1/2} b_{i'} . \end{aligned}$$

We perform Fourier transformations on a_i and $b_{i'}$:

$$a_{\mathbf{q}} = (2/N)^{1/2} \sum_i a_i \exp(i\mathbf{q} \cdot \mathbf{r}_i), \quad b_{\mathbf{q}} = (2/N)^{1/2} \sum_{i'} b_{i'} \exp(i\mathbf{q} \cdot \mathbf{r}_{i'}),$$

where $\mathbf{q}$ runs over the $N/2$ wave vectors in the Brillouin zone, referred to the chemical

unit cell. Furthermore, the following new operators, A_{qI} and A_{qII} , will be introduced:

$$\begin{aligned} A_{qI} &= (1/8)[\{(A^-/B^+)^{1/4} + (B^+/A^-)^{1/4}\}(a_q + b_q) \\ &\quad + \{(A^-/B^+)^{1/4} - (B^+/A^-)^{1/4}\}(a_q^* + b_q^*)], \\ A_{qII} &= (1/8)[\{(A^+/B^-)^{1/4} + (B^-/A^+)^{1/4}\}(a_q - b_q) \\ &\quad + \{(A^+/B^-)^{1/4} - (B^-/A^+)^{1/4}\}(a_q^* - b_q^*)], \end{aligned} \quad (56)$$

where

$$\begin{aligned} A^\pm &= 16J_1S \cos \theta (1 \pm \gamma_{qI}) - 4J_2S \cos 2\theta (1 - \gamma_{q2}) - 8J_3S (1 - \gamma_{q3}), \\ B^\pm &= 16J_1S (\cos \theta \pm \gamma_{qI}) - 4J_2S (\cos 2\theta - \gamma_{q2}) - 8J_3S (1 - \gamma_{q3}), \end{aligned} \quad (57)$$

$$\begin{aligned} \gamma_{qI} &= \cos(q_x a/2) \cos(q_y a/2) \cos(q_z c/2), \\ \gamma_{q2} &= \cos(q_z c), \quad \gamma_{q3} = \{\cos(q_x a) + \cos(q_y a)\}/2, \end{aligned} \quad (58)$$

(see (3)). Using these operators, we can derive the diagonalized spin wave Hamiltonian:

$$\begin{aligned} H_{sw} &= -N8J_1S(S+1) \cos \theta + N2J_2S(S+1) \cos 2\theta \\ &\quad + N4J_3S(S+1) + \sum_q \hbar \omega_{qI} (N_{qI} + (1/2)) + \sum_q \hbar \omega_{qII} (N_{qII} + (1/2)), \end{aligned} \quad (59)$$

where

$$\begin{aligned} N_{qI} &= A_{qI}^* A_{qI}, \quad N_{qII} = A_{qII}^* A_{qII}, \\ \hbar \omega_{qI} &= \{A^- B^+\}^{1/2}, \quad \hbar \omega_{qII} = \{A^+ B^-\}^{1/2}. \end{aligned} \quad (60)$$

As can be seen from (57), (58) and (60), ω_{qI} vanishes at $q=0$ and ω_{qII} vanishes at $q=\pm k$ (where k is the wave vector of the screw). In the neighborhood of these points, ω_{qI} and ω_{qII} can be expanded as

$$\begin{aligned} (\hbar \omega_{qI})^2 &= 8S^2 \{4J_1(1 + \cos \theta) - J_2(1 - \cos 2\theta)\} \{(J_1 \cos \theta - J_3)(q_x^2 + q_y^2)a^2 \\ &\quad + (J_1 \cos \theta - J_2 \cos 2\theta)q_z^2 c^2\}, \\ (\hbar \omega_{qII})^2 &= 8S^2 \{4J_1 \cos \theta (1 + \cos \theta) - J_2 \cos 2\theta (1 - \cos 2\theta)\} \\ &\quad \times \{(J_1 \cos \theta - J_3)(q_x^2 + q_y^2)a^2 + (J_1 \cos \theta - J_2 \cos 2\theta)(q_z \mp k)^2 c^2\}. \end{aligned} \quad (61)$$

A sketch of the variation of these frequencies against q_z is shown in Fig. 9.

The modes of oscillation of the spins in branch I at $q=0$ and branch II at $q=\pm k$, are illustrated by Fig. 10. They must be detectable by antiferromagnetic resonance at zero field, as the matrix elements of $\sum_i S_{iz} + \sum_{i'} S_{i'z}$ and $\sum_i S_{ix} + \sum_{i'} S_{i'x}$, for branch I and II respectively, do not vanish. The first mode can be excited when an oscillating field is applied along the the c -axis, and the second doubly degenerate modes by an oscillating field in the c -plane. Taking into account the crystalline anisotropy energy, (36), as well as the dipolar interaction, the frequency of the doubly degenerate modes for $q=\pm k$ is calculated to be as follows:

$$\begin{aligned} (\hbar \omega_{II})^2 &= 4S^2 \{4J_1 \cos \theta (1 + \cos \theta) - J_2 \cos 2\theta (1 - \cos 2\theta)\} \{(D/2) - (3N/8)\varphi_A(\theta)g^2\mu_B^2\}, \\ (\omega_{II}/\gamma)^2 &= 2D_1/\chi_{s//}, \end{aligned} \quad (62)$$

where $\chi_{s//}$ is given by (49), and D_1 is defined by the total anisotropy energy, E_a ,

$$E_a = D_1 \sin^2 \Theta = \{N(D'/2) - (3/2)\varphi_A(\theta)(N\mu/2)^2\} \sin^2 \Theta.$$

In the case of MnO_2 , the wavelength corresponding to the frequency of this mode is calculated to be about 200μ , assuming $16J_1 \sim 90 \text{ cm}^{-1}$, $J_2/J_1 \approx 1.60$ (see section 2) and $D \sim 90 \text{ cm}^{-1}$ (see section 4), thus it comes into the far infra-red region.

The mode of $q=0$ belonging to branch I is such that the spins oscillate as illustrated in Fig. 10 (A); the spins make a small oscillation as a whole in the c -plane, as well as out-of plane motions. Hence, its frequency is expected to be proportional to the geometrical mean between $1/\chi_{s\perp}$ and the anisotropy constant in the c -plane. The anisotropy energy in the c -plane may be very small, as discussed in section 3 (B), so that frequency of this mode may also be very low.

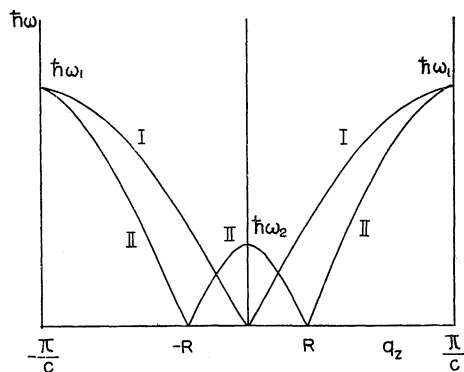


Fig. 9. Sketch of the frequency spectra.

$$\begin{aligned}\hbar\omega_1 &= S[(16J_1 \cos \theta - 8J_2 \cos \theta) \\ &\quad \{16J_1 \cos \theta - 4J_2(\cos 2\theta + 1)\}]^{1/2} \\ \hbar\omega_2 &= S[32J_1 \cos \theta \{16J_1(\cos \theta - 1) \\ &\quad + 4J_2(1 - \cos 2\theta)\}]^{1/2}\end{aligned}$$

§ 6. Neutron Diffraction Pattern

The scattering cross-section of neutrons from the screw-type structure is calculated from the standered formula⁹⁾ as

$$d\sigma/d\Omega \propto f(\kappa)^2 \{1 + (\kappa_{\perp}/\kappa)^2\} \langle S^z \rangle / S^z \{ \Delta(\kappa - \mathbf{K}) + \Delta(\kappa + \mathbf{K}) \} \sin^2 \{ ((K_1 + K_2)a + K_3c)/2 \}, \quad (63)$$

where $d\sigma/d\Omega$ is the differential cross-section, $f(\kappa)$ the form factor, κ the scattering vector, $\kappa_{\perp}$ the component of κ perpendicular to the plane to which the spins are parallel, $\mathbf{K}$ the reciprocal lattice vector referred to the chemical unit cell, K_1 , K_2 , K_3 are its components, and

Table II.

Indices	$\kappa = 4\pi \sin \theta_B / \lambda$ ($\text{\AA}^{-1}$)	$1 + (\kappa_z/\kappa)^2$	$f(\kappa)^2$ *	I/I_{005}
102	1.54	1.16	.71	.60
005	1.55	2	.69	1
115	2.54	1.38	.34	.34
009	2.80	2	.27	.39
212	3.22	1.04	.19	.14
205	3.23	1.23	.19	.17
119	3.44	1.65	.15	.18
209	3.98	1.49	.06	.06
1012	3.98	1.87	.06	.08
302	4.29	1.02	.04	.03
225	4.29	1.13	.04	.03

* The normalized form factor for Mn^{2+} (not for Mn^{4+}) by Shull et al. used. (C. G. Shull, W. A. Strauser, and E. O. Wollan: Phys. Rev. **83** (1951) 334).

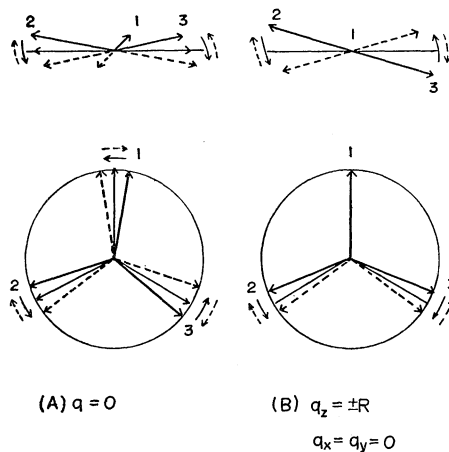


Fig. 10. The modes of the spin wave of $q=0$ in branch I (A) and $q_z=k$. ($q_x, q_y=0$) in branch II (B) (the 3-pitch-structure).

The upper figures are projections perpendicular to the c -axis. The lower figures those on the c -plane.

$$\Delta(\mathbf{X}) = (2/N) \left| \sum_i \exp(i\mathbf{X} \cdot \mathbf{r}_i) \right|^2, \quad |\mathbf{X}_x|, |\mathbf{X}_y| < \pi/a, \quad |\mathbf{X}_z| < \pi/c.$$

As easily seen from (63), the Bragg condition for the magnetic reflection is that κ is equal to $\mathbf{K} \pm \mathbf{k}$ and $\{(K_1 + K_2)a + K_3c\}/\pi$ is odd. Hence, if p is the reciprocal of the screw pitch in units of c , we expect reflections $1, 0, p, 0, 0, 1 \pm p, 1, 1, 1 \pm p, 2, 0, 1 \pm p$, etc..

Through private correspondences the writer has been informed by Prof. Erickson⁹⁾ of his experiment of neutron diffraction with a single crystal of polianite. He has observed five relations 102, 005, 115, 009, and 205, at 80°K, where his indices refer to the magnetic unit cell which is seven times large along the c -axis compared with the chemical unit cell. These reflections are just what we expect for the screw-type structure of pitch $7/2$. The calculated relative intensities of the reflections expected for the proper $7/2$ -pitch-structure are shown in Table II.

§ 7. Concluding Remarks

By a Fourier method, possible arrangements of classical vector spins in rutile-type crystal were studied, and it was found that the screw-type structure is energetically favorable under certain conditions imposed on the values of J_2/J_1 and J_3/J_1 . A similar method of "the normal mode" has been independently used by Gersch and Koehler¹⁰⁾ in their study of the spin arrangement in some simple lattices, but they assumed scalar spins which have only two discrete values, $+S$ and $-S$, so that they could not find a configuration in which the spin direction varies gradually in space.

The stability conditions for the screw-type structure with a pitch of $7/2$, which seems to be realized in polianite, are that $J_2/J_1 = 1.60$ and $J_3/J_1 < 0.31$, as described in section 2. The range of J_2/J_1 , in which the $7/2$ -pitch-structure is stabilized by the crystalline field anisotropy, is extremely narrow, as discussed in section 3. It would therefore be accidental that the pitch of the screw in polianite at 80°K be exactly $7/2$; to account for the neutron diffraction data, the pitch can be approximately $7/2$. The pitch can vary in the neighborhood of $7/2$ if J_2/J_1 varies due, for instance, to thermal expansion or an applied stress. It might be interesting to perform a neutron diffraction experiment at lower temperatures or under stress to detect such a variation in pitch.

If $J_3/J_1 > 0.31$, the $7/2$ -pitch screw-type structure cannot be realized, and the theoretical value of Θ_p/T_N cannot exceed 2.8. If we determine J_3/J_1 using Eq. (24) from the observed ratio of Θ_p (316°K) to T_N (84°K), that is, $\Theta_p/T_N = 3.8$, we obtain $J_3/J_1 \sim 0.56$. This discrepancy may partly be due to a

crude approximation of the molecular field theory.

If θ is put equal to zero in (57), (60) and (61), we have the frequencies of the spin waves in the MnF_2 -type structure. According to these frequency formulae, some modes with small $\mathbf{q}$ have imaginary frequencies when J_2 is greater than J_1 , as clearly seen from (61). This situation has been pointed out by Nagai and Nakamura¹¹⁾. However, the MnF_2 -type structure is less stable than the screw-type structure for $J_2 > J_1$. Therefore we see that, as soon as the spin superstructure becomes energetically unstable, some of the spin wave excitations become also unstable. This statement is also true when the screw-type structure becomes less stable than the B.E.Y.-structure ($2J_2J_3 > J_1^2$, $J_2/J_1 > 1$), in which case the frequencies of modes with $\mathbf{q}$ in the neighborhood of $q_x, q_y = \pm\pi/a$, $q_z = \pm\pi/c$ become imaginary. It can be shown that if only exchange energy is considered, the stability boundaries in the $J_2/J_1, J_3/J_1$ -plane derivable from the instability of the spin waves are the same with those which can be derived by comparing the energy values of the various spin superstructures. When the anisotropy energy is also taken into consideration, this becomes no longer true; for example, in the region where the MnF_2 -type structure has just become unstable, all the spin waves of the same structure remain still stable.

The writer would like to express his deep thanks to Professor T. Nagamiya of Osaka University for suggesting this study as well as his continual advice in the course of the work and to Dr. J. Kanamori of Osaka University and Mr. Y. Kitano of University of Osaka Prefecture for their helpful discussions.

He is also indebted to Professor R. A. Erickson of Ohio State University for kindly supplying his experimental data prior to publication.

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